SingleRAN

Fault Management Feature Parameter Description

Issue Draft B

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Added the statement that the function of retaining alarms uncleared following a base station reset does not take effect for alarms in ADMC mode. For details, see **3.2.2.6.10 Retaining Alarms Uncleared Following a Base Station Reset**.

Revised descriptions in this document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 07 (2025-12-14).

Technical Changes

Change Description	Parameter Change
Added the function of retaining alarms uncleared following a base station reset. For details, see **3.2.2.6.10 Retaining Alarms Uncleared Following a Base Station Reset**.	Added the **ALMFUNCTIONSW** MO.
Added support for fast alarm clearance due to the awareness of fault rectification by the RHUB and pRRU. For details, see **3.2.2.1.2 Toggling Rules**.	None
Optimized the enhanced alarm toggle filtering function. For details, see **Enhanced Alarm Toggling and Filtering** in **3.2.2.1.2 Toggling Rules**.	None
Added support for alarm reporting based on the bit error rate when the receive optical power is low. For details, see **3.2.2.6.7 Alarm Reporting Based on the Bit Error Rate When the Receive Optical Power Is Low**.	Modified parameter: Added the **LOW_RX_OP_PWR_ALM_OPT_SW** option to the **EQUIPMENT**.***EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)*** parameter.
Added support for OTDR-based fronthaul fault root cause diagnosis. For details, see **3.2.2.6.8 OTDR-based Fronthaul Fault Root Cause Diagnosis**.	Added the **DSP OPTPATHTOPO** command.
Added support for NR Cell Unavailable alarm optimization. For details, see **3.2.2.6.9 NR Cell Unavailable Alarm Optimization**.	Modified parameter: Added the **NR_CELL_UNAVAILABLE_ALM_OPT_SW** option to the **gNBOamParam**.***AlarmSwitch*** parameter.

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
SRAN	MRFD-210304	Fault Management	**3.1 Overview**
LTE FDD	LBFD-004006	Fault Management	
LTE TDD	TDLBFD-004006	Fault Management	
NB-IoT	MLBFD-12000406	Fault Management	
NR	FBFD-010025	Fault Management in the Basic O&M Package	

3 Fault Management

3.1 Overview

3.1.1 Architecture

Fault management detects and records device faults, notifies users, and recommends associated troubleshooting methods. This helps maintenance personnel quickly locate and rectify faults, minimizing the impact of faults on the network.

Fault management based on 3GPP specifications operates in the following layers:

- Network element layer (NEL)
- Element management layer (EML)
- Network management layer (NML)

Figure 3-1 shows the fault management architecture.

Figure 3-1 Fault management architecture

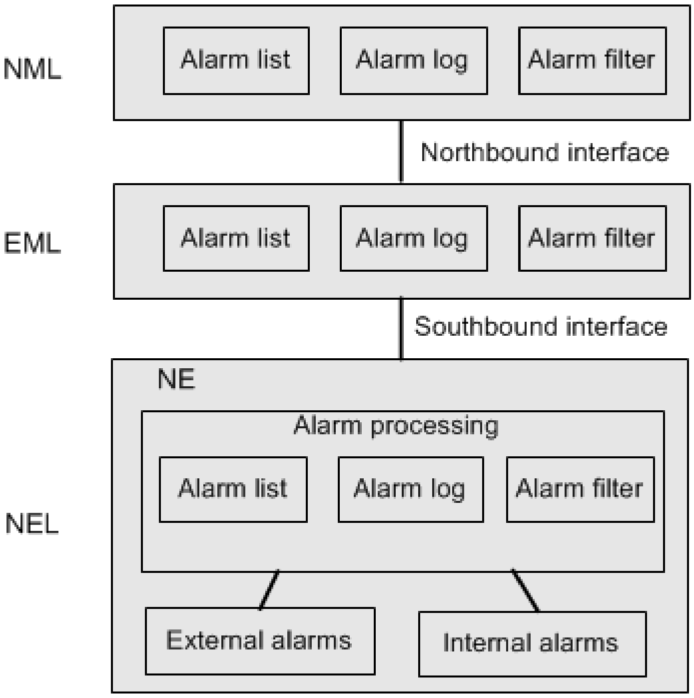


Table 3-1 describes the components in the fault management architecture.

Table 3-1 Components in the fault management architecture

Component	Description
External alarm	Handles alarms reported by peripheral devices, such as an environment monitoring device.
Internal alarm	Handles alarms reported by base station controllers and base stations.
Alarm list	Lists IDs, names, severities, and alarm types of all reported alarms.
Alarm log	Records the detailed information about each alarm.
Alarm filter	Filters alarms according to preset criteria.

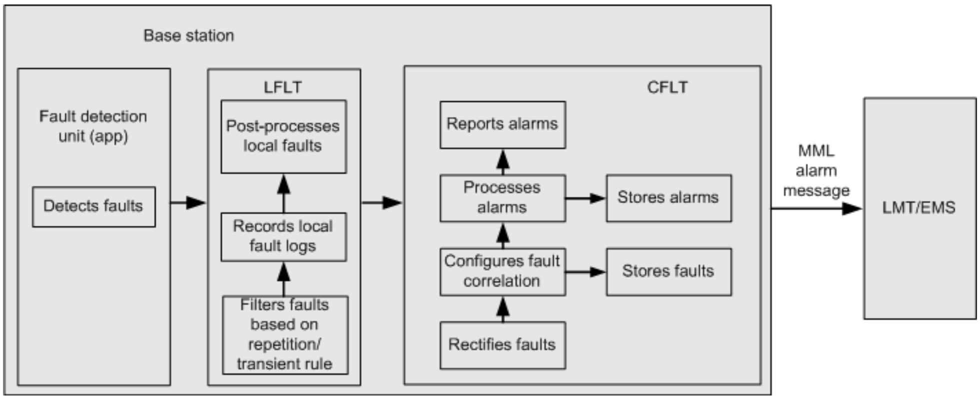
NEL

The NEL is where most alarms are generated. Most of these alarms are generated from NE devices and peripheral devices. In the integrated deployment mode, NE devices include base station controllers and base stations.

After detecting exceptions, an NE device first filters them and derives assertions based on preset rules. The exceptions that cannot be resolved are reported as faults. NE devices can automatically rectify faults. When certain faults need to be rectified with manual operations or using other automated processing equipment, alarms are reported.

Figure 3-2 shows implementation of fault management on the NEL, using Huawei wireless base stations as an example.

Figure 3-2 Fault management on the NEL



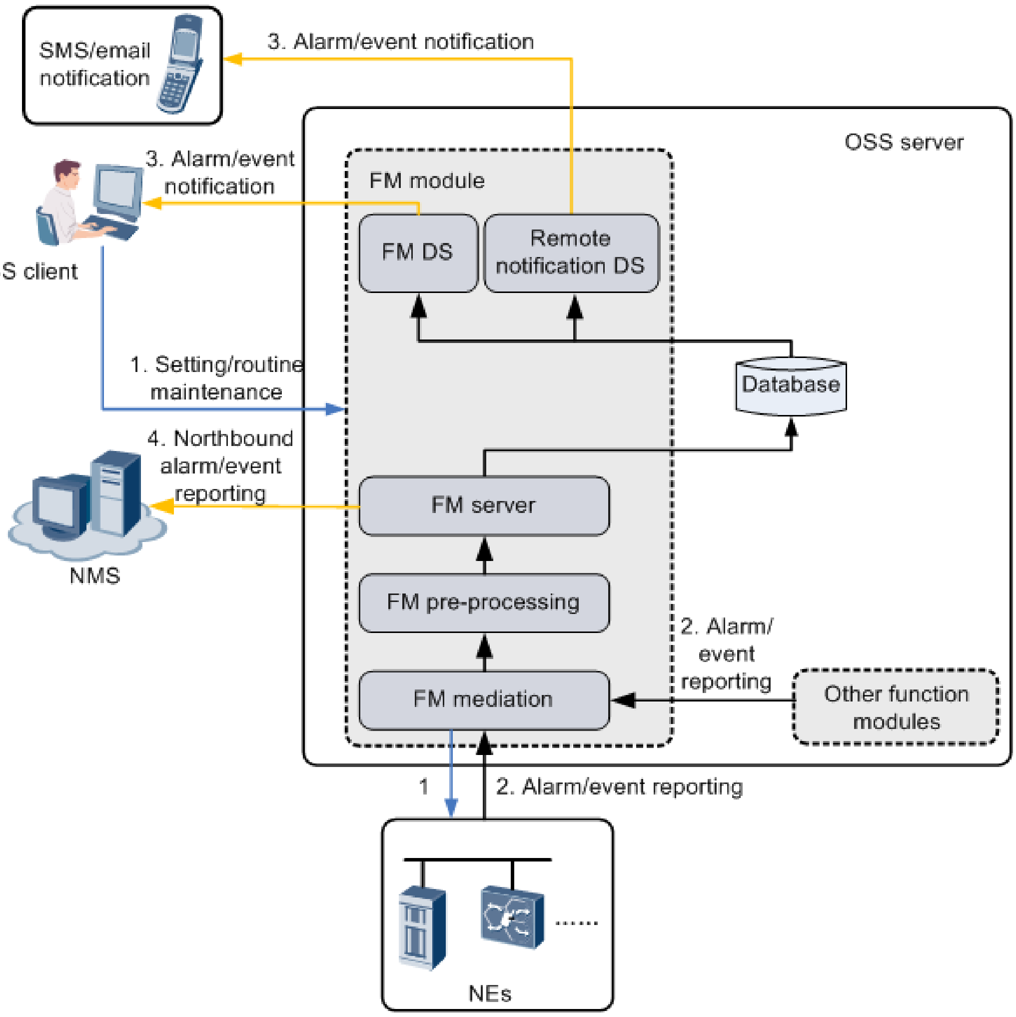
EML

A device vendor provides the EML to manage the NEs of its own. On certain EMLs, devices of multiple vendors can be managed.

On the EML, alarms are received, stored, and filtered. Alarms are dispatched through the northbound interface.

Huawei EML is the MAE. Fault management on the MAE involves alarm/event setting, alarm/event reporting, and alarm/event notification. Figure 3-3 shows the implementation of fault management on the EML, using the MAE as an example.

Figure 3-3 Fault management on the MAE



NML

Operators centrally manage their devices on the NML using a network management system (NMS). The devices are deployed on various networks, such as the radio access network (RAN), core network, and transport network. The NMS is generally developed and managed in-house by individual telecom operators. The NMS comprehensively manages the devices of different vendors and fields.

Fault management is an important function of the NMS. With this function, the NMS can receive, filter, and store alarms generated on devices of multiple vendors, dispatch work orders for these alarms, or perform other operations.

3.1.2 Basic Concepts

Table 3-2 describes the basic concepts of fault management.

Table 3-2 Basic concepts of fault management

Concept	Description	NE	EMS
Fault	A fault is a physical or logical factor that prevents a system from running properly. The fault is displayed as an error.	√	√
Alarm	An alarm is reported to the EMS when a device incurs a fault or an exception that needs to be rectified manually or using automated processing equipment. An alarm has two states: generated and cleared. When an alarm is reported, the following alarm parameters are provided: **Alarm ID**, **Alarm name**, **Alarm severity**, and **Location info**. For details about alarm parameters, see **Table 3-3**.	√	√
Event	An event refers to the information that is generated on an NE while the network is running. Such information does not indicate a fault but can be used as a reference for troubleshooting faults. Event parameters include **Event name**, **Event raised time**, and **Location info**.	√	√
Current alarm	Current alarms indicate persistent or unacknowledged alarms on the EMS side. Current alarms only apply to the EMS, since acknowledgment information is not saved on the NE side.	x	√
Active alarm	Active alarms refer to the uncleared alarms on NEs.	√	x
Duplicate alarm	Duplicate alarms refer to new alarms whose alarm types, alarm sources, key location parameters, and clearance types are the same as those of the existing alarms.	√	√
Common alarm	Common alarms refer to the alarms reported by common devices, such as power supply and temperature control devices, in multimode scenarios. Common alarms are identified by the **Common Alarm** field. The value in this field also indicates the involved RATs.	√	√

Table 3-3 describes the concepts related to alarm attributes.

Table 3-3 Concepts related to alarm attributes

Concept	Description
Alarm serial number	An alarm serial number records the sequence of alarms generated on an NE and uniquely identifies an alarm in the alarm log. The same serial number is used in alarm generation, alarm clearance, and alarm change, including the changes in alarm severity and location information (such as error codes).
Alarm ID	In a product, an alarm ID uniquely identifies an alarm and represents a specific fault or event. You can determine the details of an alarm based on its alarm ID, such as the alarm name, clearance suggestion, alarm cause, and location information. However, specific values of location information cannot be located. An alarm ID ranges from **1** to **65533**. The alarm ID of a generated alarm is the same as that of the corresponding cleared alarm.
Alarm name	In a product, an alarm name uniquely identifies an alarm and clearly and accurately indicates contents of the alarm. An alarm name maps an alarm ID. The alarm name of a generated alarm is the same as that of the corresponding cleared alarm.
Alarm severity	An alarm severity identifies how severe a fault affects services. Four alarm severities are defined in descending order: critical, major, minor, and warning. ● Critical: Faults affect system services and must be immediately rectified. If a device malfunctions or a certain type of resource becomes unavailable, immediately troubleshoot the faults (even in off-work hours). ● Major: Faults affect quality of service (QoS) and need to be immediately rectified. If the QoS of a device or resource deteriorates, troubleshoot the faults (in work hours). ● Minor: Faults may affect QoS. Troubleshoot the faults at an appropriate time or continue to observe the alarms. For example, if a packet loss alarm is reported, you need to check the settings of bit error rate (BER) thresholds, observe the onsite BER, and determine the impact. ● Warning: Faults may affect system services due to potential errors. Troubleshoot the faults based on error information. For example, the report of an alarm on insufficient redundant power supply indicates services may be affected. In this case, handle the alarm before subsequent services are affected.

Concept	Description
Alarm type	Alarms are classified by alarm source into 16 types: ● Power system: alarms for faults in the power system (providing –48 V DC power supply). ● Environmental system: alarms for faults related to the device environment, such as the temperature, humidity, and access control. ● Signaling system: alarms for faults related to signaling, such as channel associated signaling (No.1) and common channel signaling (No.7). ● Trunk system: alarms for faults on trunk circuits and trunk boards. ● Hardware system: alarms for faults on board components, such as clocks, and CPUs. ● Software system: alarms for software-related faults. ● Running system: alarms for faults that occur while a system is running. ● Communications system: alarms for faults in the communications system, such as network cable disconnection and network equipment fault. ● QoS: alarms for QoS-related faults. ● Processing error: alarms for system processing errors. ● OMC: alarms generated when the OMC is not working properly. ● Integrity: one type of security alarm, which indicates that information may be illegally modified, inserted, or deleted. ● Operations: one type of security alarm, which indicates that services are unavailable or unreachable due to incorrect operations, faults, or other unknown reasons. ● Physical system: one type of security alarm, which indicates that physical resources are damaged by suspicious hacker attacks. ● Security: one type of security alarm, which indicates that the RAN system has suffered hacker attacks. ● Time domain: one type of security alarm, which indicates that events have occurred at an unexpected or forbidden time.
Alarm log	Alarm logs record detailed information of alarms, including cleared and uncleared alarms and all events. The recorded information includes the alarm ID, alarm name, alarm severity, OSS category, alarm serial number, alarm generation time, alarm clearance time, and additional alarm information. Alarm logs do not record information of the following three types of suppressed alarms: ● Alarms whose **Shielded Flag** is set to **Shield** ● Alarms that are suppressed during alarm toggle processing ● Alarms that are suppressed during alarm correlation processing

Concept	Description
Alarm generation time	Alarm generation time marks the point when an alarm or event is generated. Alarm generation time is the current time of the module or device where an alarm is generated. For example, if an alarm is generated during local O&M, the local time is used. However, if an alarm is generated on the host, the host time is used.
Alarm clearance time	Alarm clearance time marks the point when an alarm is cleared. Typically, for cleared alarms, the alarm clearance time is the current time of the module or device where an alarm is located. For alarms cleared during local O&M, the alarm clearance time is the local time.
Notify time	Alarm notify time refers to the time when an alarm is reported on the NE side. It is different from the alarm generation time and is reported optionally by an NE to meet various service requirements.
Cleared notify time	Cleared notify time refers to the time when an alarm is cleared on the NE side. It is different from the alarm clearance time and is reported optionally by an NE to meet various service requirements.

Concept	Description
Alarm clearance type	This concept indicates how an alarm is cleared. Alarm clearance type includes: ● Normal clearance If a cleared alarm is received, an alarm is cleared. Alternatively, the OMC automatically clears the alarms that have been cleared on the NE but are not cleared on the EMS based on the active alarm list synchronized from NEs. ● Reset clearance If alarms are detected again on a device after the device restarts, old alarms are automatically cleared. ● Manual clearance If you manually clear an alarm, the alarm is displayed as a cleared alarm on the LMT even though the fault persists. Therefore, you are advised to confirm that the fault has been rectified before manually clearing an alarm. ● Configuration clearance After an object is deleted, alarms for the object are automatically cleared. ● Correlation clearance After receiving the root alarm of an uncleared correlative alarm, the fault management system automatically clears the correlative alarm when reporting the root alarm. The correlative alarm is deleted from GUIs. If alarm correlation is not configured, this alarm type is unnecessary. ● Overwrite clearance The oldest alarms are overwritten due to limited hard disk space for NEs. If active alarms are overwritten, NEs automatically clear them. ● State-switching clearance During a device status switchover, the active alarms in the previous state are automatically cleared. The alarms in the current state are reported.

Concept	Description
Additional information	Additional information is special information required by operators but is not contained in the location information specified during the alarm design. Additional information can be configured on NEs. The EMS cannot identify additional information, and only parses it by character string. Each item included in the additional information must be in the format of <*item name*=><*value*> and contains a maximum of 500 bytes. The serial number of a board is included in the alarm additional information about a board in the BBU, RHUB, RF unit, CXU, DRS, and optical module. The following restrictions apply to the board serial number: 1. When an NE is started, if a device has never been connected or started, the device serial number cannot be obtained and therefore is not contained in the alarm additional information. 2. When an NE is started, if a device has been started and then disconnected after the serial number is obtained, the alarm uses the serial number obtained before the disconnection. If the device is replaced during the disconnection and the new device has not been started after the replacement, the serial number in the alarm cannot be updated.
Location information	Location information refers to the alarm information about products and services, such as the CPU ID, board type, specific error code, and other related information used for fault troubleshooting, such as the temperature. The location information in the alarm clearance report is the same as that in the corresponding alarm report. The location information can be empty.
Alarm synchronization number	An alarm synchronization number records the sequence in which NEs report alarm messages to the EMS and ensures the synchronization between the EMS and NEs on sending and receiving alarms. Alarm synchronization numbers in an alarm generation message and the corresponding alarm clearance message are different. The alarm synchronization number ranges from **1** to **0x7ffffffe** in a cyclical order.

3.2 NE Fault Management

3.2.1 Basic Functions

Fault Management provides the following basic functions:

Fault Detection

A fault detection unit detects faults and reports them to the fault management module. The fault management module then processes the faults and reports associated alarms for these faults to the MAE or local maintenance terminal (LMT). A fault detection unit can detect faults of all managed objects (MOs), such as carriers, ports, channels, boards, base stations, cells, links, and signaling messages.

Fault Collection

This function collects the fault information reported by each fault detection unit and centrally processes this information.

Troubleshooting

Fault troubleshooting includes state changes, fault isolation, and automatic fault rectification. Base stations and base station controllers diagnose faults and automatically rectify these issues based on preset policies. If required, the preset policies can be modified by adjusting parameters, however automatic fault rectification cannot be adjusted in most circumstances. When faults fail to be automatically rectified and manual intervention is required, alarms are reported.

Fault Logs

Fault logs are classified into local fault logs and central fault logs. Local fault logs are used to analyze faults after returning faulty boards. Different from common logs, local fault logs are recorded in local non-volatile storage area. Central fault logs record the specific details about all faults, based on the obtained fault information about an NE.

Alarm Synchronization

- For MBSCs, eGBTSs, NodeBs, eNodeBs, and gNodeBs

  After an NE is created on or reconnected to the MAE, it reports alarms or events to the MAE using the alarm synchronization mechanism. If the MAE fails to receive real-time alarms/event reported by an NE due to network exceptions or other reasons, alarm/event synchronization is also required.

- For GBTSs

  The alarm synchronization between a GBTS and the MAE consists of two stages:

  – Alarm synchronization between the GBTS and the MBSC: The MBSC queries active alarms from the GBTS, issues a command to the GBTS to check for unsynchronized alarms, and updates alarm records on the MBSC based on the check result.
  – Alarm synchronization between the MBSC and the MAE.

Alarm Severity Change

The alarm severity of an uncleared alarm can be changed. After the alarm severity is changed, the alarm needs to be reported again. The alarm log contains the time when the change occurs. The serial number of the reported alarm is the same as that of the alarm when it is generated. However, the alarm synchronization number needs to be allocated again.

Common alarms

Common alarms only apply to multimode base stations. If GSM, UMTS, LTE, and NR of a multimode base station all detect a common alarm, the alarm can be displayed only on the side of a single RAT. This prevents redundant work order dispatches. The RAT in the alarm varies according to the multi-RAT priority settings.

Alarm Box Management

Alarm box management provides functions such as specifying the severity of alarms to be reported to the alarm box, resetting the alarm box, and querying the alarm box version. After you specify alarms to be reported to an alarm box, the alarm box provides audible and visual notifications for you to quickly rectify faults.

3.2.2 Advanced Functions

3.2.2.1 Alarm Filtering

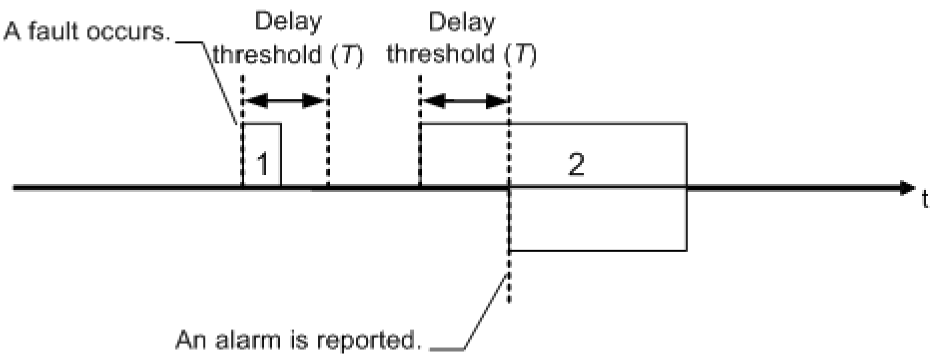
Alarm filtering involves two stages, primary filter and secondary filter. In the primary filter, fault detection units filter faults using the intermittent rules and toggling rules. In the secondary filter, alarms to be displayed to users are filtered.

3.2.2.1.1 Intermittent Rules

Faults or alarms of short duration can be filtered based on the alarm or fault generation delay. Only faults or alarms whose duration exceeds the threshold of the generation delay comply with the intermittent rules and are reserved for next stage of filtering.

As shown in **Figure 3-4**, the duration of alarm 1 is shorter than the delay threshold (*T*), so alarm 1 is discarded. The duration of alarm 2 is longer than *T*, so alarm 2 is reported.

Figure 3-4 Principles of the intermittent rules



On a base station/BSC6960, run the **SET ALMFILTER** command to set parameters related to intermittent alarm filtering.

On a base station controller, run the **SET ALMBLKSW** and **SET ALMBLKPARA** (not supported by the BSC6960) commands to set the switches and parameters related to intermittent alarm filtering:

- **SET ALMBLKSW** allows you to set **Alarm switch of blinking filter** (***BLKFILTERSW***), **Switch of statistics blinking alarm** (***BLKSTATSW***), and **Observing time window of statistical alarm** (***BLKSTATPRD***).
- **SET ALMBLKPARA** allows you to set **Alarm ID** (***AID***), **Intermittent alarm generating threshold** (***BLKPRD***), and alarm statistics thresholds. The alarm statistics thresholds include **Upper threshold for accumulated fault occurrences** (***CNTRISTHRD***), **Lower threshold for accumulated fault occurrences** (***CNTSTLTHRD***), **Upper threshold for accumulated fault duration** (***TMRISTHRD***), and **Lower threshold for accumulated fault duration** (***TMSTLTHRD***).

NOTE

To enable the intermittent alarm statistics function, both **Switch of statistics blinking alarm** (***BLKSTATSW***) and **Alarm switch of blinking filter** (***BLKFILTERSW***) must be set to **ENABLE**.

Intermittent alarm generating threshold (***BLKPRD***) must be set based on the statistics of intervals between the generation and clearance of an intermittent alarm on the live network.

In normal cases, **Observing time window of statistical alarm** (***BLKSTATPRD***), **Upper threshold for accumulated fault occurrences** (***CNTRISTHRD***), and **Lower threshold for accumulated fault occurrences** (***CNTSTLTHRD***) are set to **3600** (unit: s), **15**, and **2**, respectively.

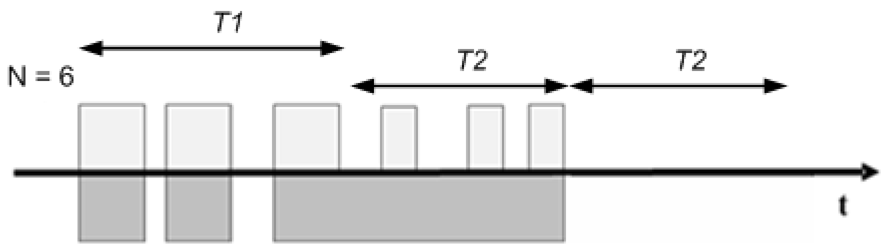
3.2.2.1.2 Toggling Rules

Manually Configured Toggling and Filtering Rules

An alarm may be repeatedly generated and cleared for an object within a certain period of time.

To resolve this issue, you can apply toggling and filtering rules to alarms that are frequently generated and cleared. **Figure 3-5** shows the principles of toggling and filtering.

Figure 3-5 Principles of toggling and filtering



Fault statistics are collected within a specific time window (*T1*). If the number of fault occurrences exceeds a certain threshold, toggling starts, the fault is considered persistent, and subsequent faults will not be reported. Once toggling begins, conditions for toggling exit are evaluated. Fault statistics are collected in another time window (*T2*, used to evaluate toggling exit). If the number of fault occurrences falls below a certain limit, the toggling ends.

On a base station, run the **SET ALMFILTER** command to set parameters related to the duration-based alarm filtering rule and times-based alarm filtering rule. The former is specified by **ALMFILTER**.*CTW* and **ALMFILTER**.*CTH*, and the latter is specified by **ALMFILTER**.*STATTW* and **ALMFILTER**.*STATRTH*.

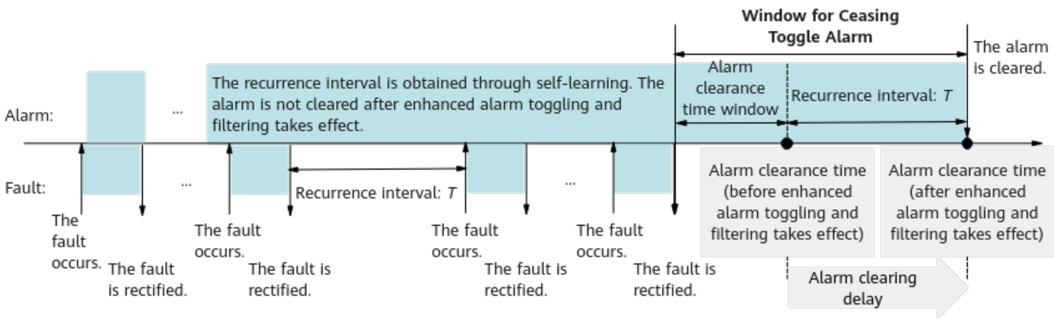
On a base station controller, run the following commands (not supported by the BSC6960) to set the switch and parameters related to alarm toggling and filtering:

- **SET ALMOSCISW** allows you to set **Alarm Oscillation Filtering Switch** (*SW*).
- **SET ALMOSCITHRD** allows you to set **Alarm ID** (*AID*), **Oscillation Entry Period** (*INOSCPRD*), **Oscillation Entry Threshold** (*INOSCTHRD*), **Oscillation Exit Period** (*OUTOSCPRD*), and **Oscillation Exit Threshold** (*OUTOSCTHRD*).

Enhanced Alarm Toggling and Filtering

Different from the preceding method of toggling and filtering through manual parameter settings, enhanced alarm toggling and filtering adaptively adjusts the toggling window based on base station faults. The system obtains the recurrence interval *T* of a fault through self-learning. *T* is then added to the alarm clearance time window. If a new fault is detected within the extended time window, the current alarm will not be cleared. By doing so, frequent alarm reporting is avoided to minimize the interference caused by massive repeated alarms to operations and maintenance. **Figure 3-6** shows the principle of enhanced alarm toggling and filtering.

Figure 3-6 Principles of enhanced alarm toggling and filtering



Alarm clearance conditions when enhanced alarm toggling and filtering is enabled:

- Before enhanced alarm toggling and filtering is enabled, an alarm can be cleared only after waiting for the alarm clearance time window once the fault is completely rectified.
- After enhanced alarm toggling and filtering is enabled, an alarm can be cleared only after waiting for the duration specified by **Window for Ceasing Toggle Alarm** once the fault is completely rectified. The duration specified by **Window for Ceasing Toggle Alarm** equals the alarm clearance time window plus the recurrence interval *T*. The alarm clearance time is extended by *T* compared with when enhanced alarm toggling and filtering is disabled. After the fault is completely rectified, the alarm may be cleared after a maximum of 30 minutes.

To enable enhanced alarm toggling and filtering for a specific alarm, run the **SET ALMTOGGLERULE** command with **ALMTOGGLERULE.***ALMTOGGLESWITCH* corresponding to the alarm ID set to **ON**. Currently, this function takes effect only for the alarms listed in **Table 3-4**.

Table 3-4 Alarms for which enhanced alarm toggling and filtering can take effect

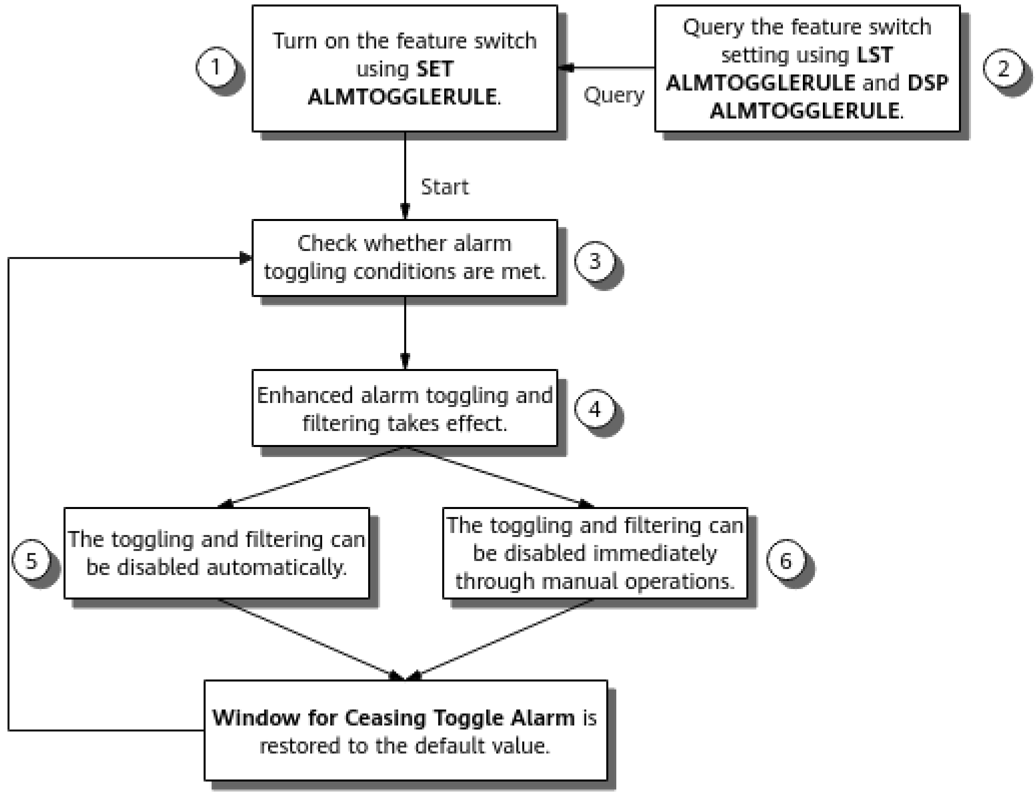
Alarm	Applicable Base Station Model
ALM-25954 User Plane Fault	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
ALM-29204 X2 Interface Fault	
ALM-29800 gNodeB X2 Interface Fault	
ALM-29810 gNodeB Xn Interface Fault	
ALM-26235 RF Unit Maintenance Link Failure	
ALM-26234 BBU CPRI Interface Error	
ALM-26504 RF Unit CPRI Interface Error	
ALM-26324 BBU IR Interface Error	3900 and 5900 series base stations (macro base stations)
ALM-26508 RF Unit IR Interface Error	

Scenarios where enhanced alarm toggling and filtering does not take effect:

- If the **SET ALMFILTER** command has been executed to set parameters related to alarm filtering, these parameter settings take effect and the parameter settings related to alarm toggling do not take effect.
- The parameter settings related to alarm toggling do not take effect for alarms in the maintenance mode.

Figure 3-7 shows the procedure of enhanced alarm toggling and filtering.

Figure 3-7 Procedure of enhanced alarm toggling and filtering



1. To turn on the feature switch for a specific alarm, run the **SET ALMTOGGLERULE** command with **ALMTOGGLERULE**.***ALMTOGGLESWITCH*** set to **ON** for the alarm based on **Alarm ID**. The alarm toggling rule will apply to the involved alarm.
2. Query the feature switch setting. Run the **LST ALMTOGGLERULE** command to obtain the setting of the alarm toggling switch, and run the **DSP ALMTOGGLERULE** command to obtain the alarm toggling status.
3. Check whether alarm toggling conditions are met. If they are, enhanced alarm toggling and filtering starts. If multiple faults map the same alarm and all are recurring, enhanced alarm toggling and filtering is more likely to be triggered.
4. Enhanced alarm toggling and filtering takes effect. The value of **Window for Ceasing Toggle Alarm** is calculated, and the value can be obtained by running the **DSP ALMTOGGLERULE** command.
5. The toggling and filtering can be disabled automatically. The alarm is cleared after the time specified by **Window for Ceasing Toggle Alarm** elapses.
6. The toggling and filtering can be disabled immediately through manual operations. Run the **DSP ALMTOGGLERULE** to obtain the alarm serial number, and then run the **OPR ALMRESTORECHK** command to reset the value of **Window for Ceasing Toggle Alarm** by the alarm serial number. When the fault is rectified, the alarm is cleared only after the time specified by **Window for Ceasing Alarm** elapses.

Impacts of enhanced alarm toggling and filtering:

- Impact on performance counters: After enhanced alarm toggling and filtering is enabled, the values of some counters may be restored to normal when an alarm is not cleared. For example, the value of **L.Sig.X2.SendSetup.Att** is restored to normal when ALM-29204 X2 Interface Fault is not cleared.

- Impact on the number of alarms:
  - If multiple alarms with the same alarm ID but different causes are repeatedly generated and cleared, alarm clearance is delayed for all of these alarms after enhanced alarm toggling and filtering is enabled. Multiple alarms may exist using the same alarm ID.
  - Alarms that are repeatedly generated and cleared are filtered to reduce the number of repeated alarms.
- Impact on alarm clearance:
  - If parameter settings related to alarm filtering are modified, such as configuring alarm masking, manually clearing an alarm will not trigger toggling and this alarm may be cleared immediately.
  - After enhanced alarm toggling and filtering is enabled, the clearance of alarms that are repeatedly generated and cleared may be delayed. Evaluate whether it is acceptable before enabling the feature. You can run the **OPR ALMRESTORECHK** command to reset the value of **Window for Ceasing Toggle Alarm** to immediately exit the feature.
- Impact on alarm correlation:
  - For an alarm that is repeatedly generated and cleared, its parent-child relationship with other alarms may not be accurately identified and subsequently these alarms cannot be correlated, causing the root and correlative alarms to be reported at the same time.
  - If the alarm toggling rule is configured for a correlative alarm but not its root alarm, the root alarm corresponding to the root serial number of the correlative alarm may be cleared first, which is normal.

Fast alarm clearance due to the awareness of fault rectification:

To reduce the impact of enhanced alarm toggling and filtering on alarm clearance, the base station automatically resets the value of **Window for Ceasing Toggle Alarm** after detecting any of the following alarm clearance operations:

- Configuration change, such as configuration object deletion and port unblocking
- Hardware change, such as the replacement, insertion/installation, or removal of an optical module, BBU, RRU, RHUB, or pRRU
- Hardware reset, such as MML-based board reset, RRU power-off reset, and RHUB/pRRU power-off reset

NOTE

- There are no standardized alarm clearance operations for customers. This function minimizes the impact on alarm clearance delay as much as possible. However, it cannot ensure that all alarm clearance operations can be accurately detected.
- To ensure that the base station can fully detect the alarm clearance operation of removing and inserting an RF optical module and prevent alarm toggling and filtering from affecting alarm clearance due to detection failures, toggling for an RF-related alarm (ALM-26235 RF Unit Maintenance Link Failure, ALM-26504 RF Unit CPRI Interface Error, or ALM-26508 RF Unit IR Interface Error) is delayed for 30 minutes after the RF optical module is removed and inserted. During this period, the alarm may be generated and cleared repeatedly.
- In separate-MPT co-SDR scenarios, fast alarm clearance due to the awareness of fault rectification takes effect only after all separate-MPT NEs are upgraded to V100R022C00 or later. For example, in CPRI MUX networking, if the base station version of the converged party is V100R022C00 or later and the base station version of the converging party is earlier than V100R022C00, the base station serving the converged party does not support this function.

Table 3-5 lists the alarms supported by this function.

Table 3-5 Alarms for which fast alarm clearance due to the awareness of fault rectification can take effect

Alarm	Applicable Base Station Model
ALM-26235 RF Unit Maintenance Link Failure	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
ALM-26234 BBU CPRI Interface Error	
ALM-26504 RF Unit CPRI Interface Error	
ALM-26324 BBU IR Interface Error	3900 and 5900 series base stations (macro base stations)
ALM-26508 RF Unit IR Interface Error	

This function has the following hardware requirements:

- Boards:
  - Main control boards: no requirements
  - Baseband processing units: only UBBP series
- RF modules:
  - For macro base stations:
    - 5000 series RRUs and 5000 series AAUs
    - Certain 3000 series RRUs and 3000 series AAUs: RRU3236E, RRU3249, RRU3260, RRU3262, RRU3268, RRU3269, RRU3276, RRU3278, RRU3278u, RRU3279, RRU3281, RRU3668, RRU3832 (except the one with the SBOM code of 02310PTJ), RRU3931E, RRU3952, RRU3952m, RRU3953, RRU3953w, RRU3954, RRU3956, RRU3956w, RRU3958, RRU3959, RRU3959a, RRU3959w, RRU3962,

AAU3910 (SBOM code: 02312DAU), AAU3911 (except the one with the SBOM code of 02311DBG), AAU3920, AAU3940, and AAU3961

- For LampSite base stations:
  - RHUBs: all RHUBs except the RHUB3908, RHUB3909, RHUB3910, and RHUB3918
  - pRRUs: all pRRUs except the pRRU3901, pRRU3902, pRRU3907, pRRU3911, pRRU3912, and pRRU3916

3.2.2.2 Alarm Correlation

Alarm correlation refers to the causal relationship between alarms indicated by the identifiers of root alarms and correlative alarms. When multiple correlative faults are triggered by a root fault, this function filters out non-root alarms and displays root alarms to users.

Related Concepts

Alarms can be correlated. Some alarms have parent and child relationships, and some have sibling relationships.

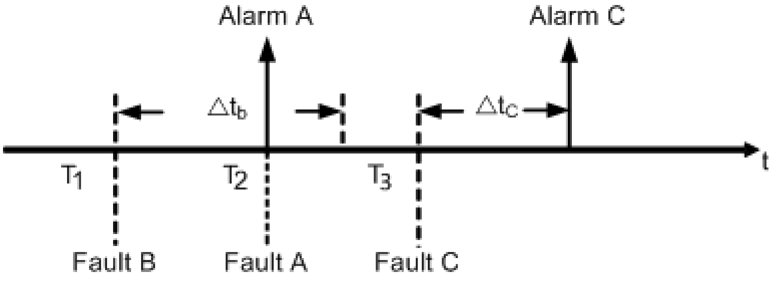
- "Parent and child" refers to the relationship between root alarms and correlative alarms. For example, when the fault that causes alarm A occurs, alarms A, B, C, and D may be simultaneously reported. If alarm A triggers alarm B, alarm B triggers alarm C, and alarm C triggers alarm D, then alarm A is the root alarm and alarms B, C, and D are its correlative alarms.
- "Sibling" refers to the relationship between alarms that are generated for the same fault. For example, when multiple optical port tributary alarms are generated on the same optical interface board, these alarms have sibling relationship with each other.

Principles

If an alarm is a correlative alarm and its root alarm can be identified by NEs, **source** of the alarm is set to **correlative alarm**. The root alarm serial number is added to the alarm message for you to efficiently locate the root alarm and rectify the fault.

Figure 3-8 is used as an example to show the principles of alarm correlation. In this example, faults B, A, and C occur in sequence and are detected at T_1, T_2, and T_3, respectively. Based on the alarm correlation rules, fault A is the root fault, fault B is a correlative fault of fault A and is suppressed, and fault C is a correlative fault of fault B and is not suppressed. The correlation analysis delays of fault B and fault C are Δt_b and Δt_c, respectively.

Figure 3-8 Examples of alarm correlation



After the alarm correlation function is enabled, alarms A, B, and C are reported as follows:

- When the root fault A is detected, the corresponding alarm A is directly reported.
- The correlation analysis of fault B is performed at T_1+Δt_b. The root fault A is detected during this time, and therefore alarm B for fault B is suppressed. If root fault A is rectified, but the correlative fault B remains uncleared, alarm B will be reported. For correlative alarm B, the reported alarm generation time is the initial fault generation time T_1 by default. If the function of refreshing alarm information is enabled by setting the **MOCRSVDPARA**.***RSVDPARAID*** parameter to **ALMRERPTTIMEFLAG** and the **MOCRSVDPARA**.***RSVDPARAVALUE*** parameter to **ON**, the alarm generation time in the reported alarm is the time when the alarm for root fault A is cleared or the time when the alarm of the last root alarm is cleared (if fault B has multiple root faults).
- The correlation analysis of fault C is performed at T_3+Δt_c. The root alarm A has been reported. In this case, alarm C, which contains the serial number of alarm A (root alarm serial number) is reported.

WARNING

The function is used to reduce the number of correlative alarms. Before using this function, you need to assess its impact.

Certain critical alarms, such as service-related alarms, cannot be suppressed based on alarm correlation. These alarms are generally generated for other faults, such as physical device faults and data transmission faults. These alarms carry the serial numbers of their root alarms. In this way, the MAE displays alarm correlations to maintenance personnel for fast fault location and troubleshooting.

WOFD-100620 Efficient Trouble Ticket Dispatching on the MAE simplifies the handling of alarms with correlations:

- If multiple root alarms or correlative alarms are involved, NEs provide a correlative alarm group ID to identify a group of correlated alarms. For example, the preceding alarms A, B, and C have the same correlative alarm group ID. The MAE can send correlative alarm group IDs to the NMS through the northbound interface, enabling operators to dispatch trouble tickets from the NMS. This improves the efficiency of trouble ticket dispatch and troubleshooting on the NMS.

- The MAE can combine multiple alarms generated for the same fault within a specific period into one alarm at the minimum.

The base station and base station controller often separately report alarms indicating the same fault (for example, a fault on the intermediate transmission device). In this scenario, you can import an inter-NE alarm correlation rule into the MAE so that correlated alarms reported by the base station and base station controller can be categorized into the same correlative alarm group. Alternatively, the correlated alarms on either the base station or base station controller side can be discarded. The predefined default correlation rules are only available after the purchase of the Efficient Trouble Ticket license.

For details, see the *Function Description About Efficient Trouble Ticket Dispatching* in *MAE Product Documentation*.

Related Configurations

The alarm correlation suppression flag can be set for certain alarms on the base station and BSC6960 to reduce the number of reported alarms. Specifically, the **SET ALMCORRSHLD** command can be executed on the base station and BSC6960 to set the **Report Flag of Son Alarm** parameter. The generated correlative alarms will not be reported when the **Report Flag of Son Alarm** parameter is set to **NOT_REPORT(Not Report)**.

If the correlative alarm needs to be cleared, run the command **RST ALMCORRSHLD** to reset the **Report Flag of Son Alarm** of the specified alarm. Before resetting, run the **LST ALMCORRSHLD** command to query all the alarms for which **Report Flag of Son Alarm** is set to **NOT_REPORT(Not Report)**.

3.2.2.3 User-Defined Alarms

Principles

Base stations and base station controllers can be connected to external environment monitoring devices to monitor the environment and device status, such as the temperature, humidity, voltage, theft, and smoke. You can define alarms on base stations and base station controllers for faults related to the status of the environment and devices. You can also set parameters for these alarms, such as the alarm name, alarm severity, and alarm type. In this way, you can dynamically monitor the environment and devices.

Before configuring user-defined alarms, you need to run the **ADD EMU** command to add an environment monitoring unit (EMU). **Table 3-6** and **Table 3-7** list the parameters for an EMU on a base station controller and a base station, respectively.

Table 3-6 Parameters for an EMU on a base station controller

Parameter Type	Parameter
Switch	● Enable Door Status Alarm Reporting (***DOOR_ENGINE_MASK***) ● Enable Humidity Alarm Reporting (***HUM_MASK***) ● Enable Infrared Sensor Alarm Reporting (***INFRA_RED_MASK***) ● Enable Smoke Alarm Reporting (***SMOKE_MASK***) ● Enable Temperature Alarm Reporting (***TEMP_MASK***) ● Enable Water Alarm Reporting (***WATER_MASK***) ● Switch for Relay 1 (***POWER_RELAY1***) to Switch for Relay 6 (***POWER_RELAY6***) ● Enable Alarm Reporting for 24V Power (***VOL24_MASK***) ● Enable Alarm Reporting for 48V Power (***VOL48_MASK***)
Threshold	● Upper Limit of Signal Output of External Analog 1 (***EX_ANO1_SIG_MAX***) to Upper Limit of Signal Output of External Analog 4 (***EX_ANO4_SIG_MAX***) ● Lower Limit of Signal Output of External Analog 1 (***EX_ANO1_SIG_MIN***) to Lower Limit of Signal Output of External Analog 4 (***EX_ANO4_SIG_MIN***) ● Upper Limit of Measurement Range of External Analog 1 (***EX_ANO1_VAL_MAX***) to Upper Limit of Measurement Range of External Analog 4 (***EX_ANO4_VAL_MAX***) ● Lower Limit of Measurement Range of External Analog 1 (***EX_ANO1_VAL_MIN***) to Lower Limit of Measurement Range of External Analog 4 (***EX_ANO4_VAL_MIN***) ● Upper Limit of Humidity Alarm (***HUM_THD_HIGH***) and Lower Limit of Humidity Alarm (***HUM_THD_LOW***) ● Upper Limit of Temperature Alarm (***TEMP_THD_HIGH***) and Lower Limit of Temperature Alarm (***TEMP_THD_LOW***) ● Upper Limit of Alarm for 24V Power (***VOL24_THD_HIGH***) and Lower Limit of Alarm for 24V Power (***VOL24_THD_LOW***) ● Upper Limit of Alarm for 48V Power (***VOL48_THD_HIGH***) and Lower Limit of Alarm for 48V Power (***VOL48_THD_LOW***)
Sensor	Sensor Type of External Analog 1 (***EX_ANO1_TYPE***) to Sensor Type of External Analog 4 (***EX_ANO4_TYPE***)

Table 3-7 Parameters for an EMU on a base station

Parameter Type	Parameter
Switch	● Special Analog Alarm Flag (**EMU.*SAAF***) ● Special Boolean Alarm Flag (**EMU.*SBAF***) ● Burglar Alarm Auto Clear Permit (**EMU.*ANTITHEFTALLOW***)
Threshold	● Temperature Alarm Lower Threshold (**EMU.*TLTHD***) and Temperature Alarm Upper Threshold (**EMU.*TUTHD***) ● Humidity Alarm Lower Threshold (**EMU.*HLTHD***) and Humidity Alarm Upper Threshold (**EMU.*HUTHD***) ● Clear Command Delay Time (**EMU.*CLRDELAY***) and Clear Command Time Interval (**EMU.*CLRINTERVAL***)

Related Configurations

To configure user-defined alarms by running commands, perform the following steps:

- On a base station
  a. Run the **SET ALMPORT** command to bind the ID of a user-defined alarm with a physical port.
  b. Run the **SET ENVALMPARA** command to set the alarm name, alarm severity, and alarm type.
- On a base station controller
  a. Run the **SET ALMPORT** command to configure the environmental signal input port for environmental alarms.
  b. Run the **SET ENVALMPARA** command to set the alarm name, alarm severity, and alarm type.

You can also configure user-defined alarms on MAE GUIs. For details, see the section "Defining an NE Alarm" of "Fault Management" in MAE product documentation.

3.2.2.4 Alarm Suppression

Principles

This function allows you to suppress unnecessary alarms by alarm ID or object.

- Suppressing alarms by alarm ID

  If **Shielded Flag** of a specified alarm ID is set to **Shielded**, all the active alarms of the alarm ID are cleared. During alarm suppression, no alarm will be reported even if the fault persists. If faults persist after alarm suppression is disabled, alarms are reported again.
- Suppressing alarms by object

You can suppress a specific alarm or all alarms for an object. If an alarm is not generated for the object, the alarm cannot be suppressed.

- For LTE and NR, you can suppress a specific alarm or all alarms for a certain board or port, or suppress a specific alarm for all boards and ports.
- For UMTS and GSM, you can suppress a specific alarm or all alarms for a specific board, port, digital signal processor (DSP), cell, or base station, or suppress a specific alarm for all these objects.

- Suppressing alarms by alarm location information

  You can specify an alarm ID by the alarm location information and set its **Shield Flag** to **Shielded**. The alarm location information must be within the scope of **Location Information** obtained by running the **LST ALMAF** command. If the alarm location information (for example, **eNodeB Function Name**) falls outside this scope, this function does not take effect.

Related Configurations

- Run the **SET ALMSHLD** command to suppress an alarm by alarm ID. Run the **LST ALMCFG** command to query the configuration of the alarm with the specified ID. The actual alarm configuration is displayed in the command output.
- Run the **ADD OBJALMSHLD** command to suppress an alarm by object. Run the **LST OBJALMSHLD** command to query the effective alarm suppression conditions on the base station. Ineffective conditions will not be displayed in the command output.
- Run the **ADD ALMSHLDRULE** command to suppress an alarm by alarm location information. Run the **LST ALMSHLDRULE** command to query the effective alarm suppression rules on the base station. Ineffective rules will not be displayed in the command output. However, all alarm suppression rules are indexed regardless of their effective status. Assume that you are adding a rule (by running the **ADD ALMSHLDRULE** command), but an error message indicating that the object already exists is displayed. If the index you reference is used by an existing ineffective alarm suppression rule, this index will not be displayed in the **LST ALMSHLDRULE** command output. In this case, you need to remove the ineffective rule with the specified index (by running the **RMV ALMSHLDRULE** command) before adding rules again.

3.2.2.5 Maintenance Mode Alarm

This section describes maintenance mode alarms, which apply to the GSM, UMTS, LTE, and NR networks.

Related Concepts

Table 3-8 describes the concepts related to maintenance mode alarms.

Table 3-8 Concepts related to maintenance mode alarms

Concept	Description
Maintenance mode	An NE being installed, commissioned, upgraded, expanded, or relocated is called an NE in engineering maintenance mode (maintenance mode for short). This mode is used to distinguish such an NE from a normal running NE (normal NE for short). It can be applied to MOs at different levels, for example, MOs at the NE and RAT levels. By default, the state of an NE in maintenance mode can be named as follows: **Install**, **Testing**, **Upgrade**, and **Expand**. The state can also be defined as needed, such as "in service". An NE is considered as being in an engineering state when it is in a non-normal state. It will enter the normal state when the engineering state ends.
NE-level maintenance mode	This mode indicates that a specific NE is in maintenance mode.
RAT-level maintenance mode	This mode indicates that a specific RAT of a co-MPT multimode base station is in maintenance mode. It is used in the following scenarios of multimode base station deployment or maintenance: ● The engineering progress differs among different RATs of a multimode base station. ● Legacy base stations need to support a new RAT. In these scenarios, the RAT-level maintenance mode can independently be set regardless of NE-level maintenance mode. **NOTE** Separate-MPT multimode base stations and single-mode base stations do not support RAT-level maintenance mode.
RF-module-level maintenance mode	RF-module-level maintenance mode is used in the following scenarios: ● A base station has been deployed and RRUs/RFUs/AAUs need to be deployed one-by-one. ● RRUs/RFUs/AAUs need to be maintained, but no unnecessary alarms need to be reported. **NOTE** RRU/RFU/AAU-level maintenance mode does not affect remote electromechanical and remote electrical tilt alarms.

Concept	Description
RHUB-level maintenance mode	RHUB-level maintenance mode is used in the following scenarios: ● A base station has been deployed and pRRUs connected to RHUBs need to be added to expand the capacity. ● Some RHUBs need to be maintained and the maintenance mode needs to be set to prevent alarms generated during maintenance operations from affecting the customer's northbound system. This function applies only to LampSite base stations. RF-module-level maintenance mode can be configured in LampSite scenarios. Alternatively, pRRUs can inherit the maintenance mode of their directly connected RHUBs, which simplifies the maintenance mode configuration.
Maintenance mode alarm	Maintenance mode alarms refer to the alarms generated after NEs, RATs, and RF modules are set to maintenance mode. Maintenance mode alarms are automatically cleared after maintenance operations have been completed. Maintenance mode alarms are handled by engineering personnel, requiring no trouble tickets. Maintenance personnel can directly view maintenance mode alarms on the MAE by customizing the display policies, or on the LMT.

Principles

After the maintenance mode management function is enabled, NEs identify the data generated in different modes and report it to the MAE based on settings. The MAE uses the appropriate policies for displaying and reporting alarms based on identifiers carried with the data. By default, maintenance mode alarms are not displayed on the MAE or reported to the centralized monitoring system of operators.

NE-level maintenance mode is applied in topology management, performance management, and alarm management.

- Topology management

  You can filter NEs in a topology view by NE status. When an NE in maintenance mode generates an alarm, the color of the NE icon in the topology view does not change.
- Performance management

  When an NE, RAT, or RF module is in an engineering state, **Reliability** of the performance result is displayed as **Unreliable**. Except for this difference, the mechanism for handling NEs in an engineering state and NEs in the normal state are the same.
- Alarm management

  Alarms generated when NEs, RATs, and RF modules are in maintenance mode are tagged as maintenance mode alarms. When the maintenance mode changes, the reported active alarms are cleared and then reported again with flags indicating the new maintenance mode.

By default, the MAE does not monitor maintenance mode alarms. To monitor these alarms, set the filter criteria in the **Advanced** dialog box. If the MAE does not monitor maintenance mode alarms, it does not display the received maintenance mode alarms, provide audible or visual alarm notifications, or forward alarms to the NMS.

This allows alarm monitoring engineers to concentrate on normal-state alarms in general circumstances, and monitor maintenance mode alarms in specific scenarios.

RAT-level maintenance mode management is only used in alarm management.

For details about maintenance mode alarms, see *Introduction to Maintenance Mode Management* in the *MAE Product Documentation*. For definition of Node alarms and Function alarms, see *3900 & 5900 Series Base Station Alarm Reference*.

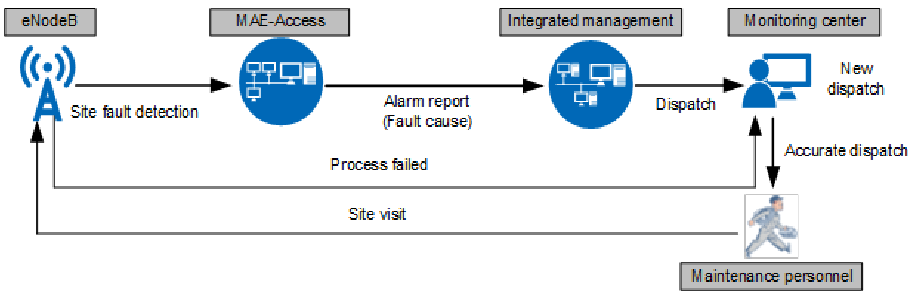
3.2.2.6 Other Functions

3.2.2.6.1 Base Station Disconnection Fault Identification

Background

30% of base station disconnection faults on live networks are caused by power and transmission faults. As shown in **Figure 3-9**, when the OSS detects that the base station, taking the eNodeB as an example, is disconnected, maintenance personnel must visit faulty sites to locate the fault. However, a large number of unnecessary work orders are dispatched since the fault cause cannot be accurately identified, wasting maintenance resources. For example, there is no need for site visits by maintenance personnel when the disconnection is caused by mains fault or transmission faults. Transmission engineers must visit the site to handle the problem.

Figure 3-9 Procedure for reporting and handling the eNodeB disconnection faults



Base station disconnection fault identification determines the causes of the base station disconnection, such as power supply fault based on the status of the faulty base station and its surrounding sites before the disconnection occurs. The alarms reported to the user by base stations of different RATs vary. The work order dispatching system receives the corresponding alarm and implements accurate dispatching based on the fault cause in the alarm to reduce unnecessary site visits.

- NodeB: ALM-22214 NodeB Unavailable
- eNodeB/gNodeB: ALM-301 NE Is Disconnected or ALM-40012 NE Is Disconnected

Principle

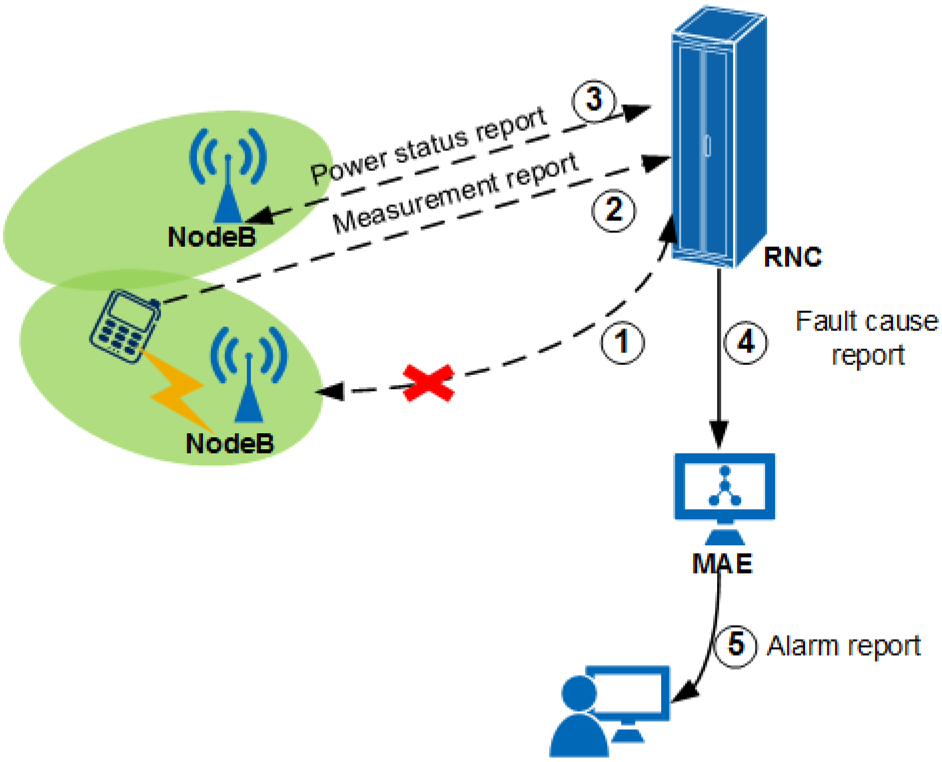
NodeB disconnection fault accurate identification mainly helps to determine whether the disconnection is due to power faults or transmission faults. "Transmission fault", "Single site power off", and "Mains fault" are added to the cause of the ALM-22214 NodeB Unavailable alarm on the RNC side. eNodeB and gNodeB disconnection fault accurate identification mainly helps to determine whether the disconnection is due to power faults or transmission faults. "Transmission interruption" and "Power supply state may be abnormal" are added to the cause of the ALM-301 NE Is Disconnected or ALM-40012 NE Is Disconnected alarm.

The cause identification principles are as follows:

- Transmission fault

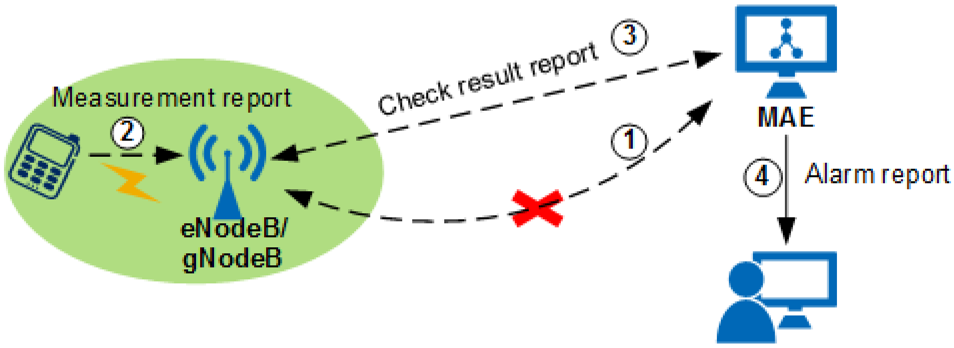
  As shown in **Figure 3-10**, when a NodeB is disconnected from an RNC and the NodeB detects a transmission fault, the transmission of the cell's pilot signals remains normal. The RNC determines whether the disconnected NodeB is powered on based on the measurement report (MR) of a UE in the neighboring cell. If the disconnected NodeB is powered on, the disconnection is caused by transmission faults.

  Figure 3-10 Transmission fault detection between the NodeB and the RNC



  As shown in **Figure 3-11**, when handshake failures between an eNodeB/gNodeB and the MAE occur for two consecutive times, the MAE finds neighboring base stations of the base station which fails to shake hands with the MAE and MR-based measurements are initiated through the neighboring base stations. The neighboring base stations determine whether the disconnected eNodeB/gNodeB is powered on based on the MRs of UEs in the neighboring cells. If the disconnected base station is powered on, the disconnection is caused by transmission faults.

Figure 3-11 Transmission fault detection between the eNodeB/gNodeB and the MAE



Transmission fault detection depends on a functioning base station for which neighbor relationships have been configured. This function cannot take effect if no neighbor relationships are configured for the cells of the disconnected base station or no surrounding base stations are functioning. In addition, if no UE access or reconfiguration takes place in the neighboring cells of the disconnected base station, MR-based measurements cannot be triggered and transmission faults cannot be detected.

- Single site power off

  The base station monitors the status of the input power with the following methods:

  – When a Huawei power monitoring unit (PMU) is configured, the power system is considered abnormal if the external AC power supply is interrupted and the standby time of the backup battery is less than 10 minutes.

  – When no Huawei PMU is configured, the BBU continuously detects the input voltage. The power system is considered abnormal if the voltage keeps decreasing to less than 51 V.

  An abnormal power system cannot continuously support base station's operation. If the base station is subsequently disconnected from the RNC or the MAE, the cause of the disconnection is identified as "Single site power off".

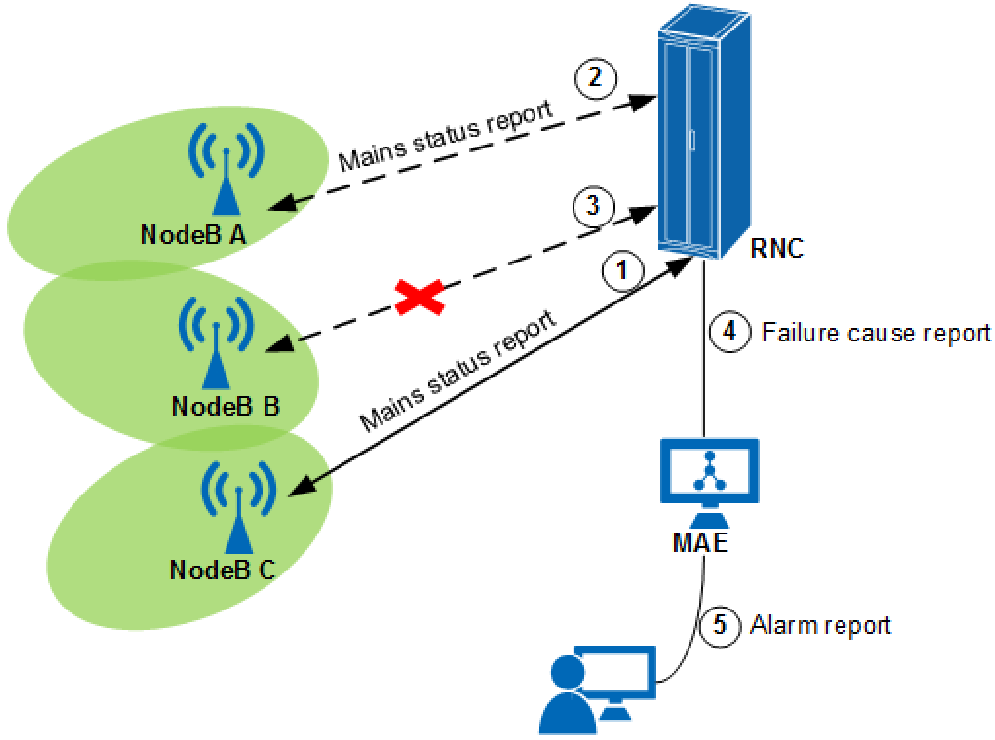
  If the base station's backup battery has severe aging issues, the base station may fail to timely report its power status to the RNC or the MAE when the voltage is abnormal or the backup battery has low standby time. As a result, the detection fails.

- Mains fault

  This function applies only to NodeBs.

  The NodeB continuously detects the status of the external AC power. As shown in **Figure 3-12**, when the NodeB detects an interruption in the input of the external AC power supply, the NodeB reports the power-off event to the RNC. The RNC records the power-off status and power-off occurrence time of each NodeB. If multiple NodeBs within a specific region are powered off at the same time, the mains is faulty.

Figure 3-12 Mains fault detection



Mains fault detection is based on the real-time detection of the status of the external AC power supply. To ensure mains fault detection when the NodeB uses a third-party power system, the third-party power system must be able to provide dry contact alarm signals for mains supply cutoff, and report the alarm to the environment monitoring signal input port on the base station. When the NodeB detects that the corresponding port generates an alarm, the external AC power supply is considered interrupted.

NOTE

The base station disconnection fault identification function cannot identify the exact cause of all base station disconnections. When the cause cannot be identified, alarms are reported based on original principles. In this case, users' maintenance personnel must visit faulty sites to ensure timely fault rectification.

Related Configurations

- On the UMTS side, run the **SET NODEBALGPARA** command to specify **NODEBALGPARA**.***SHUTDOWNFAULTDETECTSW***.
  - If this parameter is set to **ON**, the NodeB disconnection fault identification function is enabled.
  - If this parameter is set to **OFF**, the NodeB disconnection fault identification function is disabled, and the RNC reports the ALM-22214 NodeB Unavailable alarm based on the original principles.
- On the LTE side, run the **MOD GLOBALPROCSWITCH** command and set **GlobalProcSwitch**.***S1FaultCellDeactWaitTime*** to **3** to enable the eNodeB disconnection fault identification function.
- On the NR side, no configuration is required. The gNodeB disconnection fault identification function automatically takes effect.

3.2.2.6.2 Alarm Reporting in Lossless Cell Blocking Scenarios

For LTE, the **LOSSLESS_BLOCK_ALM_SW** option of the **eNodeBAlmCfg**.***AlarmSwitch*** parameter can be used to control whether to report ALM-29245 Cell Blocked when a cell is manually blocked in a lossless manner.

- If this option is selected, ALM-29245 Cell Blocked is reported when a cell is blocked in a lossless manner.
- If this option is deselected, ALM-29245 Cell Blocked is not reported when a cell is blocked in a lossless manner.

3.2.2.6.3 Board Maintenance Link Test

If ALM-26205 BBU Board Maintenance Link Failure is reported during board replacement, run the **STR BRDLNKTST** command to start a board maintenance link test. If the test result is **Fault**, the board hardware is faulty.

3.2.2.6.4 Fault Reporting Mode

In NR, if all of the cells served by a gNodeB and working in the same duplex mode are faulty, the **gNBOamParam**.***FaultRptMod*** parameter can be used to specify whether fault reporting is performed at the base station or cell level.

- If this parameter is set to **GNB_LEV**, fault reporting is performed at the base station level. When the alarm reporting conditions are met, ALM-29840 gNodeB Out of Service is reported.
- If this parameter is set to **CELL_LEV**, fault reporting is performed at the cell level. When the alarm reporting conditions are met, ALM-29841 NR Cell Unavailable is reported but ALM-29840 gNodeB Out of Service is not reported.

3.2.2.6.5 Suppression of FPGA-Fault-Triggered Resets

The "suppression of FPGA-fault-triggered resets" function can be used to determine whether the base station performs a self-healing reset if the base station detects that the clock source it uses for time synchronization is abnormal due to an FPGA soft failure. (FPGA is short for field programmable gate array.)

- With this function enabled, the base station does not perform a self-healing reset when detecting an FPGA soft failure and experiencing a clock fault. However, ALM-26200 Board Hardware Fault may be reported.
- With this function disabled, the base station performs a self-healing reset when detecting an FPGA soft failure and experiencing a clock fault. If the fault persists after the reset, ALM-26200 Board Hardware Fault may still be reported.

This function is controlled by the **FPGA_FLT_RST_SUPP_SW** option of the **TASM**.***CLKALGOPTLFEATOPTSW (LTE eNodeB, 5G gNodeB)*** parameter. It is recommended that this function be enabled only on FDD networks when the impact of clock-related exceptions on services is acceptable. This function is not recommended in other scenarios.

3.2.2.6.6 User-Plane Alarm Correlation Enhancement

In NR, if an Xn interface is faulty and shares a physical link with an eXn interface, both ALM-25954 User Plane Fault with the service type of eXn and ALM-29810 gNodeB Xn Interface Fault are reported, leading to alarm redundancy. To address this issue and improve O&M efficiency, user-plane alarm correlation enhancement is introduced.

When user-plane alarm correlation enhancement is enabled in the preceding scenario and the user plane of the eXn interface and the control plane of the Xn interface correspond to the same peer, only ALM-29810 gNodeB Xn Interface Fault is reported, while ALM-25954 User Plane Fault with the service type of eXn is not reported. The user plane of the eXn interface and the control plane of the Xn interface correspond to the same peer when both the **USERPLANEPEER**.***USERLABEL (LTE eNodeB, 5G gNodeB)*** parameter for the eXn interface and the **SCTPPEER**.***USERLABEL (LTE eNodeB, 5G gNodeB)*** parameter for the Xn interface are set to the same value in the format of *GlobalgNBId_MCC-MNC-gNodeBId*.

This function is controlled by the **TRANS_RES_LEVEL6_CHK_SW** option of the **TRANSALGEXTPARA**.***TRANSRESCHKSW (LTE eNodeB, 5G gNodeB)*** parameter. In addition, this function takes effect only when the **BEARER_OF_XN_SHIELD_SW** option of the **gNodeBParam**.***ServInterfaceAlmConfigSw*** parameter is selected in scenarios where the eXn and Xn interfaces share an endpoint group (that is, the **GNBDUEXN**.***UpEpGroupId*** and **gNBCUXn**.***UpEpGroupId*** parameters are set to the same value).

3.2.2.6.7 Alarm Reporting Based on the Bit Error Rate When the Receive Optical Power Is Low

One cause of ALM-26234 BBU CPRI Interface Error and ALM-26504 RF Unit CPRI Interface Error is that the receive power is too low, that is, the receive optical power of the optical module is too low. If one of them is reported due to too low receive power and no bit errors occur, services may not be affected. Under this circumstance, the function of alarm reporting based on the bit error rate when the receive optical power is low is introduced.

After this function is enabled, these alarms are reported only when the receive optical power of the optical module on the CPRI port is too low and bit errors occur. That is, the alarms are reported only when the receive optical power is too low and services are affected. This function can be enabled if you do not want to handle alarms that do not affect services. It is enabled by running the **SET EQUIPMENT** command with the **LOW_RX_OP_PWR_ALM_OPT_SW** option of the **EQUIPMENT**.***EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)*** parameter selected, and takes effect only when the **NEMNT**.***MNTMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **NORMAL** or **UPGRADE**.

This function takes effect only for ALM-26234 BBU CPRI Interface Error and ALM-26504 RF Unit CPRI Interface Error with the value of **Alarm Severity** being **Minor**.

This function has the following hardware requirements:

- BBUs: non-integrated BBUs
- Boards:
  - Main control boards: no requirements
  - Baseband processing units: only UBBP series
- RF modules:

– 5000 series RRUs and 5000 series AAUs
– Certain 3000 series RRUs and 3000 series AAUs: RRU3236E, RRU3249, RRU3260, RRU3262, RRU3268, RRU3269, RRU3276, RRU3278, RRU3278u, RRU3279, RRU3281, RRU3668, RRU3832 (except the one with the SBOM code of 02310PTJ), RRU3931E, RRU3952, RRU3952m, RRU3953, RRU3953w, RRU3954, RRU3956, RRU3956w, RRU3958, RRU3959, RRU3959a, RRU3959w, RRU3962, AAU3910 (SBOM code: 02312DAU), AAU3911 (except the one with the SBOM code of 02311DBG), AAU3920, AAU3940, and AAU3961

This function also requires that the value of **Protocol Type** in the **LST EQUIPMENT** command output be **CPRI**.

NOTE

- In extreme scenarios where the bit error rate is low and bit errors occur intermittently, the involved alarms will be frequently reported and cleared. As a result, it is recommended that this function be disabled in these scenarios to avoid frequent alarm reporting and clearance.
- The bit error rate can be evaluated using the **VS.RADIOEQM.CPRI0.UERR (LTE eNodeB, 5G gNodeB)** or **VS.ECPRIPORT.PCS.ErrBlocks (LTE eNodeB, 5G gNodeB)** counter. Due to the real-time requirement of alarm detection, this function determines whether bit errors occur based on the minute-level bit error rate, whereas the performance counters measure the bit error rate in a longer period. Therefore, the values of these performance counters can only be used for reference and cannot be used to match the alarm generation time and clearance time.

3.2.2.6.8 OTDR-based Fronthaul Fault Root Cause Diagnosis

With OTDR-based fronthaul fault root cause diagnosis in effect, an OTDR detection is performed and a fronthaul fault root cause alarm is reported based on the event location detected by the OTDR when the CPRI port of the OTDR optical module is faulty and services are unavailable (OTDR is short for optical time-domain reflectometer). During an OTDR detection, an OTDR optical module transmits specific pulse signals to the optical fiber, which then returns signals. By analyzing the Rayleigh scattering and Fresnel reflection of these returned signals, the OTDR optical module can determine the physical characteristics (such as optical fiber length and signal loss) of the given optical path. An algorithm then processes the data to accurately locate optical path faults and identify their causes. Optical path detection will interrupt ongoing fronthaul services because the specific pulse signals are incompatible with service signals on fronthaul optical paths.

The process of OTDR-based fronthaul fault root cause diagnosis is as follows:

1. Building fronthaul optical path topology reference information

   The base station performs optical path quality detection to generate fronthaul optical path topology reference information, which can be displayed by running the **DSP OPTPATHTOPO** command. If the value of **Topology Trust Flag** is **Reliable** in the command output, fronthaul optical path topology reference information exists and is reliable.

   The fronthaul optical path topology reference information will become invalid in the following scenarios, resulting in the reporting of EVT-26411 Fronthaul Optical Path Topology Reference Information Invalid Event:

   - The built fronthaul optical path topology reference information does not match the base station configuration in terms of networking, and the mismatch lasts for a certain period of time.
   - The OTDR detection result does not match the fronthaul optical path topology reference information.
   - The enhanced fronthaul diagnosis capability is disabled on the MAE.
2. Automatic fault diagnosis and alarm reporting

   An OTDR detection is performed when the CPRI port of the OTDR optical module is faulty and services are unavailable. The OTDR detection result is then compared with the fronthaul optical path topology reference information to check whether the information is reliable and consistent with the OTDR detection result. If the information is reliable and consistent with the OTDR detection result, a fronthaul fault root alarm is reported based on the detected event location and the following rules:
   - ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm is reported when the multiplexer/demultiplexer is detected as interrupted, dirty, or damaged during OTDR detection.
   - ALM-26410 Fronthaul Optical Fiber Exception Alarm is reported when the fronthaul optical fiber (either pigtail or trunk optical fiber) is detected as interrupted, dirty, damaged, bent, or spliced during OTDR detection.
3. Alarm handling and clearance

   ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410 Fronthaul Optical Fiber Exception Alarm is cleared when any of the following conditions is met:
   - All correlative fronthaul-related alarms of this alarm are cleared.
   - The OTDR detection result is inconsistent with the fronthaul optical path topology reference information.
   - The fronthaul optical path topology reference information is invalid due to unreliability.
   - OTDR-based fronthaul fault root cause diagnosis is disabled.
   - After a fronthaul fault occurs, OTDR detection is manually performed and no exceptions are detected at the location indicated by ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410 Fronthaul Optical Fiber Exception Alarm.

This function can be enabled only when both of the following conditions are met:

- On the MAE side: The MAE is upgraded to V100R026C00 or a later version. On the MAE, the enhanced fronthaul diagnosis capability sub-function of the intelligent alarm aggregation function is enabled.

  On the MAE, click **iFaultCare**, choose **Intelligent Fault Incident Management** > **Incident Settings**, and turn on the **Incident Enable** switch to enable the intelligent alarm aggregation function. Then, choose **Troubleshooting Policies** > **Diagnosis Ability Settings** and turn on the **Enhance the fronthaul diagnosis capability** switch to enable the enhanced fronthaul diagnosis capability.
- On the base station side:

  The fronthaul optical path topology reference information is built by running the **STR OPTPATHQUALITYDET** command.

NOTE

- Typically, multiple colored optical modules can be continuously deployed on the same baseband processing unit for networking. When three optical modules with wavelengths supported by the same multiplexer/demultiplexer are continuously deployed on a baseband processing unit, the base station checks whether the three ports on the baseband processing unit connected to these optical modules belong to the same colored optical networking based on the port numbers in ascending order. This check mechanism may lead to errors. For example, if the wavelength combinations of ports 0, 1, and 2 and ports 1, 2, and 3 on the baseband processing unit both meet the conditions, ports 0, 1, and 2 might be mistakenly considered as belonging to the same colored optical networking, while only ports 1, 2, and 3 actually belong to the same colored optical networking.
- This function can detect faults on optical paths and cannot detect whether the optical modules connected to the optical fibers are in position. If the optical modules connected to the optical fibers are not in position, ALM-26410 Fronthaul Optical Fiber Exception Alarm will not be reported.
- OTDR detection has an error in determining the event location. If the deviation between the event location displayed in the OTDR detection result and the event location described in the fronthaul optical path topology reference information is within the error range, the base station does not use the latest OTDR detection result to update the fronthaul optical path topology reference information, and the latest detection time is not updated.
- If the distance between the multiplexer/demultiplexer and the transmit end is within the allowed range of the transmit-end dead zone, the event closest to the transmit end may not be the multiplexer/demultiplexer. In this case:
  - In the **DSP OPTPATHTOPO** command output, the values of the following fields are **WDM or Connector**:
    - **BBU Side Node Type**, indicating the node closest to the BBU side
    - **RF Side Node Type**, indicating the node closest to the RF module side
  - If any of the preceding nodes is abnormal (in terms of either reflectivity or loss), ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm is reported.

  For details about the transmit-end dead zone, see *Intelligent Detection for Fronthaul Optical Paths*.
- If OTDR reflection signals cannot be detected after the position where abnormal loss occurs on the optical path, the loss in **OTDR Diagnostic Information** of ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410 Fronthaul Optical Fiber Exception Alarm is underestimated.
- If the loss of multiple adjacent optical path nodes near the transmit end cannot be individually evaluated, the displayed loss corresponding to the last node represents the overall loss, while the loss of other nodes is all displayed as NULL. If some optical path nodes cannot have their loss evaluated, their loss is also displayed as NULL. The loss is displayed in the **OTDR Diagnostic Information** parameter value of ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410 Fronthaul Optical Fiber Exception Alarm.
- OTDR detection can be automatically triggered by a CPRI port fault only once a day. For CPRI port faults occurring later in the day, you can run the **STR OPTPATHQUALITYDET** command to manually trigger OTDR detection and the reporting of ALM-26409 Fronthaul Multiplexer/Demultiplexer or Connector Exception Alarm or ALM-26410 Fronthaul Optical Fiber Exception Alarm.

This function has the following hardware requirements:

- Base station models: 3900 and 5900 series base stations (macro base stations)
- BBUs: BBU3900, BBU3910, BBU5900, and BBU5910

- Boards:
  - Main control boards: UMPTe and later
  - Baseband processing units: UBBPd, UBBPe (except UBBPeb), UBBPg, UBBPh, and later boards that support a CPRI port rate of 10 Gbit/s, 25 Gbit/s, or 50 Gbit/s
- Optical modules: 25 Gbit/s or 50 Gbit/s OTDR optical modules

The **DSP OPTPATHTOPO** command can be used to display causes of the failure to construct the fronthaul optical path topology reference information, as described in **Table 3-9**.

Table 3-9 Causes of the failure to construct the fronthaul optical path topology reference information

Topology Construction Failure Cause	Description
Root Cause Diagnosis Not Enabled	The enhanced fronthaul diagnosis capability sub-function of the intelligent alarm aggregation function is not enabled on the MAE.
Detection Not Started	The **STR OPTPATHQUALITYDET** command is not executed or fails to be executed.
Topology Type Not Supported	Currently, the fronthaul optical path topology reference information can be generated only in the chain or load-sharing topology.
Optical Path Topology Deleted	The optical path topology is deleted due to long-time absence of hardware such as the baseband processing unit and optical module, or reconstruction of the fronthaul optical path.
RF Module Not Configured	The OTDR optical module is inserted into a port on the baseband processing unit, but the other end of that connection is not equipped with an RF module.
Fronthaul Link Unavailable	The CPRI link status is abnormal when the **STR OPTPATHQUALITYDET** command is executed to build fronthaul optical path topology reference information.
Optical Module Wavelength Mismatched	In the colored optical networking, each pair of optical modules between the BBU and RF unit must have matching wavelengths.
Colored Optical Link Length Not Satisfied	In the colored optical networking, the length difference between every two of the three CPRI links exceeds 400 m.

Topology Construction Failure Cause	Description
Detection Capability Exceeded	● The length of the fronthaul optical path exceeds 10 km. ● In the colored optical networking, the number of connectors between two multiplexer/demultiplexer modules exceeds 7; in the gray optical networking, that number exceeds 9.
Detection Result Invalid	Fiber bends or heavily contaminated points in the fronthaul optical path result in incomplete detection of events beyond them.

3.2.2.6.9 NR Cell Unavailable Alarm Optimization

NR Cell Unavailable alarm optimization is introduced in NR to improve the accuracy of reported root cause for ALM-29841 NR Cell Unavailable caused by a faulty NR DU cell bound to the NR cell. This optimization improves O&M efficiency and is implemented as follows:

- If the **Specific Problem** of ALM-29844 NR DU Cell Unavailable is not **Other fault cause**, it is used as the **Specific Problem** of ALM-29841 NR Cell Unavailable.
- Otherwise, the **Specific Problem** of ALM-29870 NR DU Cell TRP Unavailable is used as the **Specific Problem** of ALM-29841 NR Cell Unavailable.

NR Cell Unavailable alarm optimization is controlled by the **NR_CELL_UNAVAILABLE_ALM_OPT_SW** option of the **gNBOamParam**.***AlarmSwitch*** parameter. It is recommended that this optimization be disabled for multi-TRP cells, such as combined cells and hyper cells, to avoid potential inaccuracies in the displayed **Specific Problem** of ALM-29841 NR Cell Unavailable for these cells.

3.2.2.6.10 Retaining Alarms Uncleared Following a Base Station Reset

The function of retaining alarms uncleared following a base station reset preserves alarms that are not cleared before a base station reset. By maintaining these alarms, this function reduces repeated alarm tickets. Before this function is enabled, the base station clears alarms that are not cleared before a base station reset after it is reset. Then, the base station detects and reports these alarms again. After this function is enabled, if any identical alarm is detected within 30 seconds after the EVT-26213 NE Startup event is reported following a base station reset, the alarm serial number remains unchanged from before the reset. If an alarm reported after a base station reset has the same alarm ID and location information as an uncleared alarm in the alarm log, the two alarms are regarded as identical.

This function is enabled by running the **SET ALMFUNCTIONSW** command with **ALMFUNCTIONSW**.***ALMRSTHOLDSWITCH (LTE eNodeB, 5G gNodeB)*** set to **ON**. This function takes effect once the base station is reset. If the base station is connected to the MAE, this function takes effect only after the MAE is upgraded to V100R026C10 or a later version.

When the base station is reset after this function is enabled, this function takes effect as follows before and after the MAE is connected:

- Before the MAE is connected, this function can take effect on the Web LMT.
- After the MAE is connected, this function takes effect if the MAE is upgraded to V100R026C10 or later. If the MAE is not upgraded to V100R026C10 or later, this function does not take effect, and the alarms that were retained before the MAE is connected will be cleared and reported again.

NOTE

- This function does not take effect for alarms that are switched to the maintenance mode, alarms in automatically detected and manually cleared (ADMC) mode, or user-defined alarms.
- This function mainly keeps the alarm serial number, alarm occurrence time, and alarm location information unchanged. After the base station is reset, the chage of other information of certain alarms triggers an alarm change.
- Alarm correlation requires a certain period of time after the base station is reset. Therefore, when a correlative alarm is successfully retained after the reset, **source** of the alarm may be temporarily changed to **root alarm** and then restored to **correlative alarm**.
- If the alarm log is inaccurate, the alarm generation time may be inaccurate after the function of retaining alarms uncleared following a base station reset is enabled. For example, if the alarm clearance is not logged due to board undervoltage, the alarm generation time in the alarm log is retained for the alarm retained after the base station is reset.
- If the alarm that is retained after the reset is suppressed, for example, the correlative alarm is suppressed after the base station is reset, the alarm clearance will be delayed for a maximum of several minutes.

3.2.3 Troubleshooting

3.2.3.1 Principles and Procedures

Principles

A fault can occur for several different reasons, and associated troubleshooting measures vary for each individual case. For example, you can troubleshoot faults remotely in the monitoring or equipment room, or onsite. You are advised to adopt following principles to improve fault troubleshooting efficiency:

- Analyze faults remotely in the monitoring room, and proceed to further troubleshoot these faults onsite.
- Identify fault causes. First check those with higher probabilities and then those with lower probabilities.
- Take troubleshooting measures with low impact on services and are low in cost. Then, take measures with high impact on services and are high in cost.

Procedures

When a fault occurs, you are recommended to stay calm, and analyze, locate, and troubleshoot the fault by following the list of procedures provided in this section.

- Check

  Check the following fault information:

  - Fault symptom
  - Time, place, and frequency that the fault has occurred
  - Fault impacts
  - Equipment's running status before the fault occurs
  - Whether alarms and correlative alarms are generated when the fault occurs
  - Whether indicators on a board are abnormal when the fault occurs

- Inquire

  Conduct an inquiry among users or staff who report faults and obtain details about the following information:

  - Operations performed on the equipment before a fault occurs and results of these operations

    For example, question whether certain data was modified, files were deleted, circuit boards were replaced, lightning or power failures had occurred, and improper operations were performed.

  - Measures taken after a fault occurs and results of these measures
  - Fault symptoms and the time, place, and frequency that the fault has occurred

- Find

  Analyze obtained information based on technical knowledge and identify causes of the faults with the help of the *Alarm Reference* and the *Troubleshooting Guide*.

- Locate and troubleshoot

  Locate the MO where the fault has occurred based on fault location principles. Rectify the fault by modifying data, replacing boards, or other necessary measures.

3.2.3.2 Fault Location and Troubleshooting

A number of alarms provide fault causes and information about field replaceable units (FRUs) or field manageable units (FMUs) where faults have occurred, for example, alarms for faults on boards. In this case, follow troubleshooting procedures in the corresponding alarm help documents to rectify these faults.

Specific types of alarms, such as link- and transmission-related alarms, do not provide fault causes or information about FRUs and FMUs because the faults triggering these alarms involve multiple devices and causes. In this case, you have to rely on your experience and fault location methods to locate and rectify these faults.

Common fault location methods are as follows:

- Ask the personnel on duty or query the operation logs of the faulty NEs.
- Compare the current data with the previous data.
- Reset hardware. In normal cases, hardware reset can rectify software-related faults.
- Remove and reinstall the boards or disconnect and reconnect the cables.
- Replace faulty units, such as boards and cables, with normal ones.
- Replace faulty units, such as boards and cables, with new ones.
- If the fault cannot be located, sequentially shut down or block the units one by one.
- Check whether the fault is caused by environmental factors. A number of faults can be rectified once adverse environmental factors, such as interference, are removed. Others cannot be rectified because environmental factors, such as earthquake, have inflicted permanent damage.

Table 3-10 lists the faults that trigger alarms and fault location methods.

Table 3-10 Classification of faults that trigger alarms and corresponding fault location methods

Fault	Description	Fault Location
Incorrect manual operations	Incorrect data configuration or manual maintenance	Check whether any data is modified or maintenance operations are performed manually. If yes, it is advisable to first check the modifications. ● Ask the personnel on duty and query the operation logs. ● Compare the current data with the previous data.
Software-related faults	Exceptions that occur when software is running	Reset hardware.
Hardware-related faults	Natural breakdown or a malfunction due to incorrect manual operations	● Remove and reinstall. ● Replace faulty units with normal ones. ● Replace faulty units with new ones. ● Shut down or block the units one by one.

Fault	Description	Fault Location
Faults on peripheral devices	Faults on peripheral devices, such as the environment monitoring device, transmission device, and antenna system	• Remove and reinstall. • Replace faulty units with normal ones. • Shut down or block the units one by one.
Adverse ambient environment	Electromagnetic interference, temperature, and adverse weather conditions or natural disasters, such as wind, rain, snow, hail, lightning, and earthquakes	• Ask the personnel on duty or query the operation logs. • Observe the ambient environment.

3.2.3.3 FMA

The Fault Management Assistance (FMA) allows users to quickly analyze fault causes and perform troubleshooting when network faults or major emergencies occur. For details about FMA, see *FMA Feature Parameter Description*.

The FMA provides two functions, namely wireless fault management and transmission fault management. The FMA focuses on resolving and optimizing problems related to wireless fault management and transmission fault management. It supports one-click mode, which reduces the technical skills required for troubleshooting accident-level problems onsite. With FMA, more problems can be resolved onsite, which greatly increases network or NE fault location efficiency. This reduces maintenance costs, and improves the transmission-layer quality, thereby improving user experience.

3.3 MAE Fault Management

This chapter provides an overview of the MAE fault management functions, such as alarm display and statistics, audible and visual alarm notification, alarm acknowledgment, and alarm synchronization. For details, see section "Fault Management" in the MAE online help.

Alarm Display and Statistics

The MAE receives alarms generated on NEs in real time and displays and collects statistics on the alarms in multiple ways.

- Alarm display

  The MAE displays alarms on alarm panels or in a bar chart and allows you to query alarms.

  – Alarm panel

Alarm panels collect and display alarms with different severities and status for MOs based on alarm list templates. Functioning as the monitoring panels, alarm panels provide the fault status on the entire network.

  – Alarm bar chart

    The MAE client provides alarm bar charts to display alarms. An alarm bar chart window contains one or more alarm bar charts. Alarms collected by an alarm template are displayed in an alarm bar chart using graphics and numerals. Functioning as the monitoring panels, alarm bar charts provide the fault status on the entire network.

  – Alarm query

    You can view the alarm list, and query alarm logs or event logs on the MAE. Alarms can be displayed in a list on an MAE GUI by alarm status or alarm severity.

    MAE supports alarm query and display of a single NE or of base stations serving neighboring cells of a base station.

    Alarm logs record all the alarms received by the MAE. Each alarm is displayed as a record.

    Event logs record all the events received by the MAE. Each event is displayed as a record.

    The alarm list displays the alarms that must be handled. One object may generate multiple alarms with the same information. However, the alarm list only displays the latest record.

- Alarm statistics

  The MAE can collect statistics on alarm and event logs based on preset statistical criteria. For example, the number of alarms of a specific severity that are reported by an NE per hour can be collected.

Audible and Visual Alarm Notification

The MAE uses the alarm box, audio adapter, and sound box to notify you of any alarms.

- Alarm box

  Alarm boxes provided by Huawei are supported on the MAE. When an alarm that complies with filter criteria is generated, you will receive audible and visual alarm notifications from the alarm box.

- Audio adapter and sound box

  You can configure different audio files for the alarms of different severities on the MAE client. When an alarm is reported, the MAE client where the audio adapter and sound box are installed plays corresponding sounds to notify you of different types of alarms.

Alarm Acknowledgment

An acknowledged alarm is cleared by users. If this alarm needs to be addressed again, unacknowledge this alarm and take corresponding measures to clear it.

The MAE supports manual acknowledgment and unacknowledgment, as well as automatic acknowledgment by alarm severity and user-defined rule.

Alarm Synchronization

The MAE provides automatic and manual data synchronization. In most cases, the MAE automatically synchronizes alarm data from NEs. However, issues such as network disconnection may cause inconsistency between the alarm data on the MAE and that on NEs. Manual data synchronization is supported on the MAE to ensure that the alarm status of NEs is accurate.

Alarm Clearance

Manually clear the alarms on the MAE that cannot be automatically cleared or that have been acknowledged.

Alarm Masking/Suppression

You can set the following alarm masking rules on the MAE or NE reporting alarms: (1) Masking alarms on the MAE. That is, after alarms are reported to the MAE, the MAE discards the alarms that meet the masking conditions and does not save them in the alarm database. (2) Masking alarms on NEs. That is, NEs do not report alarms that meet masking conditions to the MAE.

Conversion from Events to ADMC Alarms

The MAE allows you to convert an event into an alarm in automatically detected and manually cleared (ADMC) mode. The ADMC alarms are displayed in the alarm list and draw users' attention to the event.

Alarm Redefinition

By redefining alarms, you can change alarm names, types, and severities displayed on the MAE, highlight the alarms that require attention, and ignore any alarms deemed unimportant.

You can set redefinition rules on the MAE or NE side. Redefining alarm or event severities on the MAE: The MAE displays the redefined alarm severity of a received alarm. Redefining alarm or event severities on NEs: NEs report alarms to the MAE based on the redefined severities.

Alarm Correlation Analysis

The MAE supports the alarm/event correlation analysis, alarm/event frequency analysis, intermittent alarm analysis, duplicate event analysis, and analysis of the duration between the time when an alarm is acknowledged and the time when the alarm is cleared. The MAE discards alarms and suppresses non-root alarms that are also called correlative alarms or redefine alarm severities based on preset alarm correlation rules. Therefore, only root alarms and the alarms that require attention are displayed on the MAE.

The MAE provides the alarm correlation intelligent mining function. The MAE automatically generates alarm correlation rules after analyzing historical alarm data to associate correlated alarms.

Alarm Combination

The MAE provides two alarm/event combination functions:

- Function 1: combines multiple alarms into one alarm based on the values of key fields. This function is not under license control.
- Function 2: combines multiple alarms into fewer alarms based on the values of key fields and the alarm reporting time. The combined alarms contain key location information about each alarm. For example, after multiple optical port tributary alarms are combined into one alarm, tributary IDs for all the optical port tributary alarms are listed in the **Tributary ID** field of the alarm message. This function is controlled by the **Efficient Trouble Ticket** license.

Function 1 is different from function 2 in that function 1 is not restricted by time and does not provide alarm key location information such as tributary IDs.

Alarm Maintenance Experience

Alarm maintenance experience is recorded in the alarm experience database. Alarm maintenance experience can be imported into or exported from the alarm experience database. When a similar fault occurs, you can refer to relevant information to rectify the fault.

Remote Alarm Notification

When a generated alarm meets filter criteria, the MAE automatically notifies the specified maintenance engineers of the alarm by email or SMS to quickly help rectify faults. The notification criteria, notification time, notification method, and message format can be set on the MAE.

Alarm Auto-Triggering Script

This function allows you to set the script triggering criteria and specify the scripts to be triggered. The MAE server automatically runs the specified script if generated alarms meet the preset script triggering criteria. In this way, you can write an alarm auto-triggering script to perform the repetitive operations for routine maintenance, which implements automation of partial routine maintenance.

Alarm Customization

The MAE provides the environment monitoring function. You can define alarms to monitor the physical conditions of NEs based on site requirements.

3.4 Schemes for Deleting Alarms and Alarm Location Parameters

Alarms are generally added when new features are introduced or features in earlier versions are optimized. Adding or deleting alarms has an impact on the northbound interface. Alarms may be deleted for the following reasons:

- Service changes in a new version: Some alarms no longer apply to scenarios in the new version.

- Internal design optimization: Some alarms are no longer required for their corresponding faults. For example, NEs can now independently rectify the faults.
- Alarm optimization: Some alarms are replaced with other alarms.

In addition, alarm location parameters of an alarm may be deleted for the following reasons:

- Service optimization: Some alarm location parameters become invalid.
- Configuration model optimization: Some alarm location parameters are invalid or replaced with other parameters.

Alarms and alarm location parameters to be deleted will be reserved in two NE versions: the current version and the later version. **Table 3-11** describes the schemes for deleting alarms and alarm location parameters. In the table, *N* indicates the current version.

Table 3-11 Schemes for deleting alarms and alarm location parameters

Scenario	Scheme (in the N to N+1 Version)	Scheme (in the N+2 Version)
Alarms without application scenarios are deleted.	Alarms are not reported and are only reserved in related NE documents.	Alarms are deleted.
Old alarms are replaced with new alarms due to alarm optimization.	• Old alarms are not reported and are only reserved in related NE documents that provide the mapping between original alarms and new alarms. • New alarms are reported properly.	Old alarms are deleted.
Alarm location parameters become invalid and are deleted due to service optimization.	Alarm location parameters are reserved, and parameter values reported through the northbound interface are null values.	Alarm location parameters are deleted.
Old alarm location parameters are invalid due to configuration model optimization and are deleted or replaced with other parameters.	• Old alarm location parameters are reserved and reported through the northbound interface. The parameter value validity depends on the validity of the values for the corresponding parameters in the configuration model. • New alarm location parameters are reported properly.	Old alarm location parameters are deleted.

Table 3-12 lists the documents with information about deleted alarms and alarm location parameters.

Table 3-12 Documents with information about deleted alarms or alarm location parameters

Document	Description of Deleted Alarms and Alarm Location Parameters
Disuse Alarm List	This document is released with the NE software and describes all deleted alarms in this version. It also provides disuse statements that describe the reasons for deleting these alarms or alarm location parameters.
Disuse Event List	This document is released with the NE software and describes all deleted events in this version. It also provides disuse statements that describe the reasons for deleting these events or event location parameters.
Alarm Reference	Disuse statements that describe the schemes for deleting alarms or alarm location parameters are provided in the **Description** field. This document is integrated into the NE ICS Lite documentation package.

3.5 Related Features

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

3.6 Impact on the Network

System Capacity

None

Network Performance

After the base station shutdown fault identification function is enabled, handover-related KPIs such as **Intra-Frequency Handover Out Success Rate** and **Inter-Frequency Handover Out Success Rate** on a neighboring base station may deteriorate in scenarios where all S1 interfaces of the NE fail.

4 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

1. *BSC6900 GU Alarm Reference*
2. *BSC6910 GU Alarm Reference*
3. *BSC6900 GU Event Reference*
4. *BSC6910 GU Event Reference*
5. *3900 & 5900 Series Base Station Alarm Reference*
6. *BSC6900 GSM Alarm Reference*
7. *BSC6910 GSM Alarm Reference*
8. *BSC6900 GSM Event Reference*
9. *BSC6910 GSM Event Reference*
10. *GBSS Troubleshooting Guide*
11. *BSC6900 UMTS Alarm Reference*
12. *BSC6910 UMTS Alarm Reference*
13. *BSC6900 UMTS Event Reference*
14. *BSC6910 UMTS Event Reference*
15. *RAN Troubleshooting Guide*
16. *eRAN Troubleshooting Guide*
17. MAE *Fault Management*
18. *MAE Alarm Reference*
19. *MAE Function Description About Maintenance Mode Management*
20. *Intelligent Detection for Fronthaul Optical Paths*